

1. Administrative & Study Identification

Full Study Title: An Open-Label, Single-Dose Study to Evaluate the Pharmacokinetics of MK-5684 in Male Participants With Moderate Hepatic Impairment **Short Title/Acronym:** A Study to Evaluate Opevesostat (MK-5684) in Male Participants With Moderate Hepatic Impairment (MK-5684-009) **Protocol Number:** MK-5684-009 **NCT Number:** NCT06860243 **Posting ID:** 1005746320773528575 **Sponsor / Company Name:** Merck **Investigational Product:** Opevesostat (MK-5684) [along with hormone replacement therapy like Fludrocortisone acetate, Prednisone, and cortisone (Cortone, Cortone acetate)] **Phase of Development:** Phase 1 **Medical Monitor:** Merck Clinical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the pharmacokinetic (PK) profile of a single oral dose of opevesostat over time in male participants with moderate hepatic impairment compared to healthy matched control participants. **Secondary Objectives:** To assess the overall safety and tolerability of a single dose of opevesostat in the target populations.

2.2 Background & Rationale Opevesostat (MK-5684) is a novel CYP11A1 inhibitor being developed as a new way to treat castration-resistant prostate cancer by targeting the rate-limiting enzyme in androgen synthesis. Because the liver plays a critical role in drug metabolism, it is necessary to conduct a dedicated pharmacokinetic study. The rationale is to determine how moderate hepatic (liver) impairment affects the body's ability to absorb, metabolize, and excrete opevesostat compared to individuals with normal hepatic function, which will ultimately guide safe dosing recommendations.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Pharmacokinetic Study) **Control Type:** Parallel Group (Healthy matched participants) **Blinding:** Open-label **Randomization:** N/A (Non-randomized assignment based on hepatic function)

3.2 Planned Enrollment & Duration Target \$N\$: 16 evaluable male participants (distributed between the healthy control group and the moderate hepatic impairment group) **Number of Sites:** Selected specialized Phase 1 clinical pharmacology units **Trial Status:**

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male participants, aged ≥ 18 years. 2. Continuous non-smoker or moderate smoker (≤ 10 cigarettes per day or equivalent) for at least 3 months prior to dosing. 3. Body mass index (BMI) ≥ 18.0 and ≤ 42.0 kg/m². 4. For the moderate hepatic impairment cohort: Has a diagnosis of chronic, stable, hepatic insufficiency with features of cirrhosis due to any etiology.

4.2 Exclusion Criteria 1. Has a history of cancer (malignancy) or positive results for human immunodeficiency virus (HIV), hepatitis B virus (HBV), or hepatitis C virus (HCV). 2. Has a first-degree relative with multiple unexplained syncopal events, unexplained cardiac arrest, or sudden cardiac death, or has a known family history of an inherited arrhythmia syndrome (including Brugada syndrome). 3. For participants with moderate hepatic impairment: Has unstable electrolyte abnormalities (or abnormalities considered difficult to manage) or has a history of liver or other solid organ transplantation.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: A single oral dose of opevesostat (MK-5684). **Route of Administration:** Oral **Duration of Treatment:** Single administration (1 day), followed by PK tracking over approximately 4 days.

5.2 Concomitant & Prohibited Therapy Permitted: Participants will receive hormone replacement therapy alongside opevesostat to manage the systemic effects of CYP11A1 inhibition. This may include Fludrocortisone acetate, Prednisone, or cortisone (Trade names: Cortone, Cortone acetate; ATC Codes: H02AB10, S01BA03). Allowed maintenance medications for stable hepatic impairment. **Prohibited:** CYP3A4 strong inducers/inhibitors or any drugs known to heavily impact hepatic clearance that could confound the PK readings.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Days -14 to -1	Informed Consent, Medical History, BMI, Viral screening, ECG, Hepatic function staging
Baseline	Day 1	In-clinic admission, Single oral dose administration of MK-5684 + Hormone therapy
Interim	Days 1 to 4	Intensive Pharmacokinetic (PK) monitoring via

serial blood sampling | | **End of Study** | Follow-up (~Week 2) |
Final Safety Labs, Vital Signs, Exit Interview |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: Pharmacokinetic profile of opevesostat evaluated primarily by the following measurements: 1. Area Under the Concentration Versus Time Curve from 0 to Infinity (AUC_{0-inf}) of Opevesostat. 2. Area Under the Concentration Versus Time Curve from 0 to the Last Quantifiable Sample (AUC_{0-last}) of Opevesostat. 3. Area Under the Concentration Versus Time Curve from 0 to 24 hours (AUC₀₋₂₄) of Opevesostat. **Measurement Method:** Plasma samples analyzed via validated liquid chromatography-mass spectrometry (LC-MS/MS) taken at multiple protocol-defined timepoints.

7.2 Secondary & Exploratory Endpoints 1. Number of Participants Who Experience One or More Adverse Events (AEs). 2. Number of Participants Who Discontinue Study Due to an AE. Vital sign alterations and electrocardiogram (ECG) parameter shifts from baseline are also assessed.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous inpatient monitoring during the Day 1-4 stay, including physical exams, ECG monitoring for arrhythmias, and vital signs. **Serious Adverse Events (SAEs):** Investigator reporting timeline of 24 hours to the sponsor. **Data Monitoring Committee (DMC):** Internal safety review by the Sponsor's clinical study team.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Set (all subjects who received the study drug and provided sufficient concentration data) and Safety Set (all participants who received at least one dose). **Sample Size Justification:** A targeted enrollment of 16 participants provides adequate precision for estimating the ratio of geometric means for PK parameters between impaired and healthy groups. **Handling of Missing Data:** Concentrations below the lower limit of quantification (LLOQ) treated as zero for descriptive statistics; missing PK data generally excluded from non-compartmental analysis rather than imputed.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Direct onsite monitoring during the inpatient confinement period to ensure protocol adherence and strict timing of the PK sample draws. **Audit Trail:** Validated 21 CFR Part 11 compliant Electronic Data Capture (EDC) system for eCRF with full audit trails.

11. Study Identifiers & Governance

Primary ID: MK-5684-009 **Registry Number:** NCT06860243 **Posting ID:** 1005746320773528575 **Source Level:** 5 **Sponsor/Lead Organization:** Merck **Company:** Merck **Study Title:** A Study to Evaluate Opevesostat (MK-5684) in Male Participants With Moderate Hepatic Impairment (MK-5684-009) **Official Regulatory Title:** An Open-Label, Single-Dose Study to Evaluate the Pharmacokinetics of MK-5684 in Male Participants With Moderate Hepatic Impairment

12. Clinical Framework (Hepatic Specific)

Disease Areas: Hepatic Impairment, Healthy Participants **Primary Condition / Preferred Name:** Hepatic impairment (UMLS Preferred Name: Moderate Hepatic Impairment) **Semantic Type:** Disease or Syndrome **Condition Definition:** Hepatic impairment refers to a reduction in the normal functional capacity of the liver. This condition arises from various forms of liver injury, disease, or dysfunction, which compromise its crucial roles in metabolism, detoxification, protein synthesis, and bile production. Consequently, individuals may experience a range of systemic effects, including jaundice, fatigue, fluid retention, impaired drug metabolism, and altered coagulation, depending on the severity and underlying cause. **Terminology Codes:** MeSH Code: D008107 | SNOMED ID: 416629002 **Medical Methodology:** Pharmacokinetic comparative bridging **Optimization Target:** Pharmacokinetic parameter definition (AUCs) to derive dosing guidelines.

13. Study Procedures & Methodology

Management Pathway: Intensive Phase 1 clinical pharmacology unit pathway. **Education Protocol (HCP):** Site initiation visits and detailed laboratory manual training for exact PK blood draw and processing times. **Education Protocol (Patient):** Strict guidance on fasting, dietary restrictions, and smoking limitations (non-smokers or ≤ 10 cigarettes/day). **Quality Audit Mechanism:** Independent bioanalytical laboratory audits for assay validation and incurred sample reanalysis (ISR).

14. Observation & Follow-Up Schedule

Total Duration: Approximately 3-4 weeks from screening to final follow-up. **Onsite Visit Cadence:** Continuous confinement for ~4 days post-dose. **Remote Monitoring Frequency:** N/A (Direct observation phase 1 study). **Checklist Review Interval:** Daily during the inpatient portion.

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					